

A user defined software specification for FISHGUARD AI

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FISHGUARD AI is an imaginary company that produces buoyancy vehicles used for underwater data acquisition in fish farms and is based in Stavanger, Norway.

CCS Concepts: • **Software and its engineering** → **Requirements analysis**; **Software design engineering**; **Software design techniques**; **Object oriented development**; **Software prototyping**.

Godspower Ogaga Uti. 2026. A user defined software specification for FISHGUARD AI. 1, 1 (January 2026), 1 page. <https://usn.instructure.com/courses/36038/assignments/131048>

1 Background

Today, fish farming in Norway is prolific along most of the country’s coastline, producing more than 1,5 million tonnes of fish per year, almost all of which is exported. In 2017, Norway accounted for over 52% of the world’s Atlantic salmon production. Mortality levels are high in Norwegian fish farms. In 2023, 37.7 million farmed salmon smolts died, along with 62.8 million salmon during the seawater phase. Altogether, 100.5 million farmed salmon died that year. The major causes of fish death include gill disease, winter ulcers, and intense handling, such as during delousing procedures[Dyrevern 2024]. As part of a proactive measure, FISHGUARD AI developed a buoyancy vehicle for underwater data acquisition. The device has a camera and an embedded IoT system running an edge image recognition machine learning algorithm to detect sick fish in a fish farm, helping farmers with early intervention. The current system also has sensors that measure temperature, pressure, and other aquatic conditions of the farm. The device uses a TF(TransFlash) memory card to store collected data. At the end of each dive, farmers can extract data from the device. The device uses Wi-Fi and Bluetooth for external connectivity. However, many farmers struggle to make meaningful decisions from the overall data collected. Due to growing market demand for this device and client requests, the directors of FISHGUARD AI have decided to develop an optional software dashboard for their clients. This software dashboard will be subscription-based and will provide the following functionalities:

- Collect and store data from the underwater vehicle
- Remote access and control of the underwater vehicle
- Predictive analytics for risk assessment
- Automated recommendations

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2 User interaction

FISHGUARD AI maintains a list of its clients along with their respective issued device numbers. The company has a website, which they intend to extend to accommodate the new information system. After registering on the site, clients can browse services, select items, and add them to a virtual cart. At the end of the tour, clients can buy what is in their cart,remove items, or quit without making a purchase. Payment is made by submitting credit/debit card details. Upon a successful payment confirmation, the client is granted access to the purchased service(s).

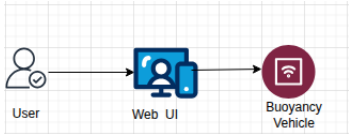


Fig. 1. Interaction Flow.

3 Data flow

3.1 Input of Information

- Users: Login details, control commands, configuration changes.
- Under water device: Sensor values, device status.

3.2 Output of information

- Users: Real-time data, alerts, reports.
- Database: Stored sensor readings and logs.
- Under water device: Control commands.

3.3 Handling of Configuration Data

Only authorized users can modify configuration settings, and changes are applied securely and dynamically.

- Device information (device ID, type).
- User preferences and alert thresholds.

References

Dyrevern. 2024. *Fish farming in Norway*. Retrieved January 19, 2026 from <https://dyrevern.no/dyrevern/fish-farming-in-norway/>

Received 27 January 2026